

Author: Sudip Gurung

Date: June 12, 2026

Abstract:

This project started as a personal challenge. I wanted to see if I could build a machine learning model that detects phishing messages, not just in English but in Nepali as well. I created a small bilingual dataset (110 samples: 60 SMS, 50 URLs) and experimented with feature engineering, TF-IDF and three classifiers (Logistic Regression, Random Forest, XGBoost). The results surprise me 100% accuracy on SMS with Logistic Regression and XGBoost and 100% on URLs with all three models. I didn't expected for models to work well with both English and Nepali texts. This project taught me that phishing has a universal pattern across languages and that even a small carefully built dataset can produce strong results. I am sharing this as a part of my machine learning portfolio to show my approach to real-world problems, feature engineering and bilingual NLP.

1. Introduction:

I do not have any cybersecurity background. I actually was into web development or UI/UX design but due to increase in popularity of machine learning and topics like AI, it attracted me.

Phishing attacks are no longer just an English language problem. In Nepal, scams targeting popular digital wallets like eSewa and khalti now appear in both English and Nepali, and increasingly, through phone calls, SMS and fake customer support messages.

The official eSewa warning notice itself lists several common baits that fraudsters use. For example:

- "Hi Sir/Ma'am, you have won an exclusive offer reserved for our top customers only. Please share your SMS Detail Send By eSewa so we can transfer the gift money to your eSewa Account."
- "Hi, I'm calling from eSewa. We have launched a KYC drive across Nepal. Doing KYC would mean you can accept up to NRs 1 lakh in your eSewa Wallet. However, I would need some proof of identification and your details."
- "Congratulations! You have been nominated as a 'Top eSewa Merchant' in your locality and are now eligible for a Special Booster Pack."

These examples reveal a clear pattern: fake rewards, fake KYC requirements and fake account upgrades. The goal is always the same, pressure the user into sharing sensitive information or clicking a malicious link. According to eSewa's advisory, fraudsters can even use technology to mask their real phone numbers, making a scam call appear as the legitimate eSewa customer care phone number (1660-01-02121).

This project started as a personal learning exercise for my machine learning portfolio. I wanted to challenge myself with a real-world problem that three specific limitations:

1. **Bilingual input:** The model had to handle both English and Nepali text without separate pipelines. Most phishing detection systems are built for one language only, English. If we give them a Nepali language, they ignore/misclassify/break entirely.
2. **Small Data:** I would not have thousands of samples, so feature engineering would matter more than throwing data at a deep learning model. I manually created 16 hand crafted features for SMS and 20 features for URLs. These feature capture the logic of phishing regardless of specific wording so, even simple models like logistic regression could perform well.
3. **Two content types:** SMS messages and URLs have very different structures, so treating them the same way would not work. SMS messages being a sentence or two of natural language and URL is a structured string (exception to natural word boundaries and grammar).

The goal was never to build a production-ready system instead, I wanted to document my process: building a dataset from scratch based on real scam patterns (with examples like eSewa). This serves as a portfolio case study to show how I approach a problem from data creation to evaluation.

2. Related Work:

A lot of research has been done on phishing detection using machine learning. Most of it focuses on English emails or general English web content. Studies using TF-IDF with logistic regression or random forests are common and they work well but only for English.

There's also work on URL based phishing detection using features like URL length, number of dots and suspicious TLDs (.xyz, .info, etc) but what's missing is work that combines SMS and URL data in a bilingual context, especially for Nepali. This paper tries to fill that small gap.

3.0. Methodology:

3.0.1. Data construction:

I created a dataset from scratch because I could not find a public bilingual Nepali-English phishing dataset. The final dataset has 110 samples:

- **SMS(60 samples):** 15 English(Legitimate/Phishing) and 15 Nepali(Legitimate/Phishing)
- **URLs (50 samples):** 25(Legitimate/Phishing) URLs (from real Nepali banking and wallet sites like esewa.com.np, khalti.com).

The phishing SMS examples included pressure tactics, fake rewards and urgency words in both language.

3.0.2. Preprocessing and Feature Engineering: For SMS, I did two things. First, I cleaned the text by lowercasing, removing punctuation and replacing sensitive patterns like URLs, phone numbers, transaction IDs and currency amounts with placeholders (e.g PHISHING_URL, MONEY_AMT). That way the model learns patterns not exact strings.

Second I build hand-crafted features based on phishing indicators:

- Urgency words (urgent, immediately, turanta)
- Threat words (blocked, suspended, banda)
- Reward words (free, winner, cashback)
- Presence of suspicious TLDs, IP addresses, HTTP(not HTTPS)

For URLs, I extracted features like URL length, number of dots/hyphens, presence of HTTPS, suspicious TLDs and whether the URL contained words like "verify", "login", "secure" or "update".

3.0.3. Model Training:

I split the data 80/20 and tested three models on SMS and three on URLs:

- Logistic Regression
- Random Forest(100 estimators for SMS, 1000 for URLs)
- XGBoost

For SMS, I combined TF-IDF feature (unigrams + bigrams) with the hand-crafted features. For URLs, I only used the numerical feature set because URLs are short and don't benefit much from TF-IDF.

4.0. Results:

The results came out better than I expected.

4.0.1. SMS Detection:

- Logistic Regression: 100% accuracy, 1.0 AUC-ROC
- XGBoost: 100% accuracy, 1.0 AUC-ROC
- Random Forest: 83.3% accuracy, 0.9722 AUC-ROC

4.0.2. URL Detection:

- All three models (Logistic Regression, Random Forest and XGBoost) scored 100% across the board.

4.0.3. Language-specific check: The best SMS model (Logistic Regression) got 100% accuracy in both English and Nepali test samples. That means the bilingual approach did not hurt performance. In fact, it worked perfectly on this dataset.

5. Discussion

Everything worked perfectly because my dataset was small and clean. In Real-world data is messier and another reason: phishing messages regardless of language, share a common structure.

Phishing URLs targets Nepali wallets often use clearly fake domains like *esewa-verify-np.com* or *khalti-secure.info*. These are very different from the legitimate *.com.np* domains.

That being said there are limitations. 110 samples is not enough to generalize to the real world. In practice, new phishing campaigns will use domains and wording that the model has never seen. The perfect scores here are promising but not proof of real-world readiness.

6. Limitations and Future Work:

1. **Small Dataset:** This is the biggest issue. I plan to collect more real-world SMS and URL samples over the next few months, possibly using crowdsourcing or working with local cybersecurity groups.
2. **Class Balance:** The dataset is perfectly balanced (50/50), which is not realistic. Real phishing is rare compared to legitimate traffic. Future work should use imbalanced data and test with metrics like precision-recall curves.
3. **No Deep Learning:** I didn't use transformers or language models (like multilingual BERT) because of the small dataset size. That could be a next step once more data is available.

7. Conclusion:

This research showed that bilingual phishing detection for Nepali and English is very feasible. Even with a small dataset, careful feature engineering especially combining TF-IDF with hand-crafted phishing indicators, produced perfect or near-perfect results. For users in Nepal, a tool like this could be built into messaging apps or mobile wallets to warn people before they click. The next step is scaling up the data and testing in the wild.